A deficit law for Strong Pillar 1: removing u tail factors costs exactly u levels of sign-kill

(the off–hook family $\lambda = (2, 2, 1^m)$, uniformly in m)

Clio

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Abstract

The strong per–subset operator vanishing $M(S) \equiv 0$ on V^λ for $|S| < D = \tau(\tau+1)/2$ governs the off–hook order law $\operatorname{ord}_{x=q^2} Z_\lambda = D$. Via the linear–combination identity of the May 28 reduction paper, the per–subset chain $M_T(S_T)$ is a signed sum of *factor–removed* tail chains $\Omega_{\text{tail}}^{(\lambda),[\widehat{U}]}$, so per–subset Pillar 1 reduces to the uniform *Strong Pillar 1* statement on the $\Omega_{\text{tail}}^{(\lambda),[\widehat{U}]}$. We prove, uniformly in m, the sharp *deficit form* of Strong Pillar 1 for the off–hook family $\lambda = (2, 2, 1^m)$:

$$\Omega_{\text{tail}}^{(\lambda),[\widehat{U}]} V^\lambda \subseteq \bigcap_{j=1}^{\tau-|U|} E_j^-\big|_{V^\lambda} \qquad \text{for every } U \subseteq \text{tail}, \ |U| < \tau,$$

i.e. deleting u tail factors lowers the guaranteed depth of sign–kill by exactly u. The engine is an elementary structural fact about the contributor hook $(2, 1^m)$: on the hook seminormal module *every* generator $S_i = T_i + 1$ acts as a rank–one operator whose image is supported on the two basis tableaux with arm value in $\{i, i+1\}$. Hence a descending product $\prod_{i\in A} S_i$ has image pinned by its leftmost factor $S_{\max A}$, with support arm value $\geq \max A \geq |A|$; this gives the hook deficit bound with no induction. The off–hook bound follows by the established R'_n–branching (only contributor: the hook $(2, 1^m)$; no vanishers): when the leftmost tail generator T_{n-1} survives, the image lands in $E^+_{n-1} = \Phi(V^{(2,1^m)})$ and pulls through to the hook bound; when T_{n-1} is deleted, the chain lies in $H_q(S_{n-1})$ and branches to the larger hook $(2, 1^{m+1})$ (closed again by the rank–one fact) and the smaller off–hook $(2, 2, 1^{m-1})$ (closed by induction on m). As a corollary we obtain refined per–subset Pillar 1 $M_T(S_T)V^\lambda \subseteq \bigcap_{j=1}^{\tau-|S_T|} E_j^-$ for all $|S_T| < \tau$ and all m, upgrading the May 28 result (which proved only 2 of 11 single–factor positions, at $m = 3$ only) to a uniform theorem.

1 Setup

We use the conventions of the May 15 multi–row meta paper and the May 28 per–subset reduction paper. $H_n(q)$ is the Iwahori–Hecke algebra with $T_i^2 = (q-1)T_i + q$; on the Hoefsmit seminormal module V^λ we set $S_i := T_i + 1$ (so $S_i^2 = (q+1)S_i$),

$$E_i^+ := \ker(T_i - q)|_{V^\lambda} = \operatorname{im}(S_i|_{V^\lambda}), \qquad E_i^- := \ker(S_i)|_{V^\lambda},$$

and $R'_k := S_{k-1}S_{k-2}\cdots S_1$. For a partition $\nu \vdash N$ with $\ell = \ell(\nu)$, the tail chain is $\Omega_{\text{tail}}^{(\nu)} := R'_{\ell+1}R'_{\ell+2}\cdots R'_N$.

Prefix length and the support rule. For a standard tableau T put $p(T) := \max\{P \geq 0 : T(i,1) = i\ \forall 1 \leq i \leq P\}$, the length of the identity prefix down column 1. For a subspace W, $\operatorname{supp}(W) := \{T : \exists\, w \in W,\ [v_T]w \neq 0\}$. We use repeatedly the *support rule* (May 14): if every $T \in \operatorname{supp}(W)$ has $p(T) \geq P$, then $W \subseteq \bigcap_{j=1}^{P-1} E_j^-$ (because entries $j, j+1$ then sit in column 1 at rows $j, j+1$, hence in the same column, forcing the (-1)–eigenvalue of T_j).

The off–hook family. For $\lambda = (2,2,1^m)$ one has $\widehat{\lambda} = (2,2)$, $\ell(\widehat{\lambda}) = 2$, $n = m+4$, $\ell := \ell(\lambda) = m+2$, and

$$\tau := \tau(\lambda) = m-1, \qquad D := \tfrac{\tau(\tau+1)}{2} = \binom{m}{2}.$$

Because $\widehat{\lambda} = (2,2)$ is fixed, the Pillar 1 split $\Omega = R \cdot \Omega^{(\lambda)}_{\text{tail}}$ with $R = R'_2 \cdots R'_\ell$ has tail consisting of *exactly two blocks*,

$$\Omega^{(\lambda)}_{\text{tail}} = R'_{n-1}\, R'_n, \qquad R'_{n-1} = S_{n-2} \cdots S_1, \quad R'_n = S_{n-1} \cdots S_1.$$

The tail letters occupy word positions $\binom{\ell}{2}, \ldots, N-1$; a tail subset S_T selects which of these letters carry P_{-1} rather than P_q.

Factor–removed chains and the reduction identity. For $U \subseteq$ tail, $\Omega^{(\lambda),[\widehat{U}]}_{\text{tail}}$ is the product of the tail S–factors *not* in U, in the left–to–right staircase order. The May 28 identity reads

$$(q+1)^{|\text{tail}|}\, M_T(S_T) = \sum_{U \subseteq S_T} (-1)^{|S_T|-|U|} (q+1)^{|U|}\, \Omega^{(\lambda),[\widehat{U}]}_{\text{tail}}, \tag{1}$$

where $M_T(S_T) = \prod_{k \in \text{tail}} A_k$, $A_k = P_{-1}^{(i_k)}$ for $k \in S_T$ and $P_q^{(i_k)}$ otherwise. The *refined Strong Pillar 1* statement is

$$\Omega^{(\lambda),[\widehat{U}]}_{\text{tail}}\, V^\lambda \ \subseteq\ \bigcap_{j=1}^{\tau-|U|} E_j^-\Big|_{V^\lambda} \qquad (|U| < \tau), \tag{2}$$

and (1) shows at once that (2) for all $U \subseteq S_T$ implies refined per–subset Pillar 1 $M_T(S_T)V^\lambda \subseteq \bigcap_{j=1}^{\tau-|S_T|} E_j^-$ for $|S_T| < \tau$. Proving (2) uniformly in m is the goal of this paper.

2 The contributor hook $(2,1^m)$ is rank–one

Let $\mu = (2,1^m)$, a hook with $|\mu| = m+2$, generators $T_1, \ldots, T_{m+1}$, and $\tau(\mu) = m-1$ (the hook column–1 threshold). Its standard tableaux are indexed by the *arm value* $a := T(1,2) \in \{2,3,\ldots,m+2\}$: cell $(1,1)$ always holds 1, the arm cell $(1,2)$ holds a, and the remaining values $\{2,\ldots,m+2\} \setminus \{a\}$ fill column 1 down rows $2,3,\ldots,m+1$ in increasing order. Write v_a for the seminormal basis vector of the tableau with arm value a; thus $\dim V^\mu = m+1$ and

$$p(v_a) = a-1 \tag{3}$$

(the column reads $1,2,\ldots,a-1$ before the arm value displaces the count).

Lemma 1 (Each generator is rank one on the hook)**.** *For $1 \leq i \leq m+1$, the operator $S_i = T_i + 1$ on V^μ has rank 1, and*

$$\operatorname{im}(S_i|_{V^\mu}) = \langle w_i \rangle, \qquad w_i \in \langle v_i, v_{i+1} \rangle \quad (i \geq 2), \qquad w_1 \in \langle v_2 \rangle.$$

In particular $\operatorname{supp}(\operatorname{im} S_i) \subseteq \{\, a = i,\ a = i+1 \,\}$.

Proof. Fix i and consider how T_i acts on v_a in the seminormal form, which is dictated by the relative position of the entries $i, i+1$ in the tableau with arm value a.

Case $a \notin \{i, i+1\}$. Then neither i nor $i+1$ is the arm value, so both lie in column 1 (recall 1 is always in column 1 as well). Entries in the same column give the (-1)–eigenvalue: $T_i v_a = -v_a$, hence $S_i v_a = 0$.

Case $i = 1$. If $a = 2$ then $1, 2$ occupy $(1,1), (1,2)$, the same *row*, so $T_1 v_2 = q v_2$ and $S_1 v_2 = (q+1) v_2$; if $a \geq 3$ then $1, 2$ are both in column 1 and $S_1 v_a = 0$. Thus $\operatorname{im}(S_1) = \langle v_2 \rangle$, rank one, $w_1 = v_2$.

Case $i \geq 2$ *and* $a \in \{i, i+1\}$. Exactly the two tableaux v_i, v_{i+1} have $i, i+1$ in neither the same row nor the same column (one of $i, i+1$ is the arm value, the other a column–1 cell; the two cells share no row and no column). On the two–dimensional space $\langle v_i, v_{i+1} \rangle$ the generator T_i acts by the standard Hoefsmit 2×2 block, whose eigenvalues are q and -1 (an honest 2–block, since the axial distance is $\neq \pm 1$). Hence $S_i = T_i + 1$ has eigenvalues $q+1$ and 0 there, i.e. rank one, with image the q–eigenline $\langle w_i \rangle \subseteq \langle v_i, v_{i+1} \rangle$.

Combining the cases: S_i kills every v_a with $a \notin \{i, i+1\}$ and has rank one on the two–dimensional complement, so $\operatorname{rank}(S_i|_{V^\mu}) = 1$ and $\operatorname{im}(S_i) = \langle w_i \rangle$ with $\operatorname{supp}(w_i) \subseteq \{i, i+1\}$. □

Proposition 2 (Hook deficit bound). *Let $A \subseteq \{1, \ldots, m+1\}$ be nonempty and let $\Pi_A := \prod_{i \in A}^{\downarrow} S_i = S_{a_r} S_{a_{r-1}} \cdots S_{a_1}$ denote the product in* decreasing *order of the elements $a_1 < \cdots < a_r$ of A. Then either $\Pi_A|_{V^\mu} = 0$, or*

$$\operatorname{im}(\Pi_A|_{V^\mu}) = \operatorname{im}(S_{\max A}|_{V^\mu}) = \langle w_{\max A} \rangle, \qquad \operatorname{supp}(\operatorname{im} \Pi_A) \subseteq \{\, a = \max A,\ a = \max A + 1 \,\}.$$

Consequently every $T \in \operatorname{supp}(\Pi_A V^\mu)$ has $p(T) \geq \max A - 1 \geq |A| - 1$, and

$$\Pi_A \, V^\mu \ \subseteq \ \bigcap_{j=1}^{\max A - 2} E_j^- \big|_{V^\mu} \ \subseteq \ \bigcap_{j=1}^{|A|-2} E_j^- \big|_{V^\mu}. \tag{4}$$

Proof. The leftmost factor of Π_A is $S_{a_r} = S_{\max A}$, so $\operatorname{im}(\Pi_A) \subseteq \operatorname{im}(S_{\max A}) = \langle w_{\max A} \rangle$ by Lemma 1. As $\langle w_{\max A} \rangle$ is one–dimensional, the image is either 0 or all of it. In either case $\operatorname{supp}(\operatorname{im} \Pi_A) \subseteq \operatorname{supp}(w_{\max A}) \subseteq \{\max A, \max A + 1\}$. By (3), any T in this support has arm value $a \geq \max A$, hence $p(T) = a - 1 \geq \max A - 1$. Since the elements of A are distinct positive integers, $\max A \geq |A|$, giving $p(T) \geq |A| - 1$. The containment (4) is the support rule applied with $P = \max A - 1$ (resp. $|A| - 1$). □

Corollary 3 (Hook Strong Pillar 1, deficit form). *For the hook $\mu = (2, 1^m)$, $\tau(\mu) = m - 1$, write the tail as the single block $\Omega^{(\mu)}_{\text{tail}} = R'_{m+2} = S_{m+1} \cdots S_1$. For any $U \subseteq \{$tail factors$\}$ with $u := |U|$,*

$$(R'_{m+2})^{[\widehat{U}]} \, V^\mu \ \subseteq \ \bigcap_{j=1}^{\tau(\mu) - u} E_j^- \big|_{V^\mu}.$$

The bound is sharp: deleting the u leftmost factors gives $(R'_{m+2})^{[\widehat{U}]} = R'_{m+2-u} = \Pi_{\{1, \ldots, m+1-u\}}$, whose image has support arm value $= m + 1 - u$, realising depth exactly $\tau(\mu) - u$.

Proof. $(R'_{m+2})^{[\widehat{U}]} = \Pi_A$ with $A = \{1, \ldots, m+1\} \setminus (\text{letters of } U)$, $|A| = m + 1 - u$. By Proposition 2, depth $\geq |A| - 2 = m - 1 - u = \tau(\mu) - u$. Sharpness: deleting the u leftmost factors leaves $A = \{1, \ldots, m + 1 - u\}$, $\max A = m + 1 - u$, support arm value $m + 1 - u$, $p = m - u$, depth $= m - 1 - u$. □

3 Lifting to the off–hook family

We now prove (2) for $\lambda = (2,2,1^m)$. The branching engine is the established reduction (May 15 meta paper, May 28 reduction paper) specialised to $\widehat{\lambda} = (2,2)$: the removable corners of λ are $c = (2,2)$ and $c_* = (m+2,1)$, so $R'_n V^\lambda = E^+_{n-1} = \Phi_{2,*}(V^{(2,1^m)})$ is a *single contributor* Φ–image of the hook $\mu = (2,1^m)$, and there are *no vanisher pieces* (no second corner inside $\widehat{\lambda}$, no admissible same–row pair). Here $\Phi := \Phi_{2,*}$ is the T_i–equivariant ($i \le n-3$) seminormal injection placing $\{n-1,n\}$ at $\{c,c_*\}$, and it preserves the column–1 prefix (its added cells lie in column ≥ 2 or strictly below the prefix region). We freely use:

- *Pull–through* (Step 1 of the reduction theorem): for $w \in V^\mu$, $R'_{n-1}\,\Phi(w) = (T_{n-2}+1)\,\Phi\big(R'^{(\mu)}_{m+2}\,w\big)$, and more generally a factor deleted from R'_{n-1} pulls through to the correspondingly deleted factor of the inner block $R'^{(\mu)}_{m+2}$ (the equivariance is factor–by–factor).
- *Prefix preservation*: Φ does not decrease $p(\cdot)$, and the single outer factor $(T_{n-2}+1)$ preserves the prefix bound (outer–factor preservation, May 14), since $n-2 = \ell \ge p+1$ in the stable regime.

Fix $U \subseteq$ tail with $u = |U| < \tau$, and split $U = U_{n-1} \sqcup U_n$ across the two blocks. Let $g := T_{n-1}$, the unique leftmost factor of R'_n.

Case A/B2: the leftmost tail generator T_{n-1} survives ($g \notin U_n$)

Lemma 4. *If $g \notin U_n$ then $\Omega^{(\lambda),[\widehat{U}]}_{\text{tail}} V^\lambda \subseteq \bigcap_{j=1}^{\tau-|U_{n-1}|} E^-_j \subseteq \bigcap_{j=1}^{\tau-u} E^-_j$.*

Proof. Write $R'^{[\widehat{U_n}]}_n = S_{n-1}\cdot G$ with $G \in H_q(S_{n-1})$ (the surviving factors of R'_n below S_{n-1}, a product over generators $\le n-2$). Then $\operatorname{im}\big(R'^{[\widehat{U_n}]}_n\big) \subseteq \operatorname{im}(S_{n-1}) = E^+_{n-1} = \Phi(V^\mu)$, so $R'^{[\widehat{U_n}]}_n V^\lambda = \Phi(W')$ for some $W' \subseteq V^\mu$. Applying the first (gapped) block and pulling through Φ,

$$\Omega^{(\lambda),[\widehat{U}]}_{\text{tail}} V^\lambda = R'^{[\widehat{U_{n-1}}]}_{n-1}\,\Phi(W') = (T_{n-2}+1)\,\Phi\big((R'^{(\mu)}_{m+2})^{[\widehat{U'_{n-1}}]}W'\big) \subseteq (T_{n-2}+1)\,\Phi\big((R'^{(\mu)}_{m+2})^{[\widehat{U'_{n-1}}]}V^\mu\big),$$

where U'_{n-1} is the image of U_{n-1} under the factor–by–factor pull–through (the equivariant factors are S_i, $i \le n-3$; the single factor S_{n-2} is the outer one). If $S_{n-2} \notin U_{n-1}$ the outer factor $(T_{n-2}+1)$ is present and $|U'_{n-1}| = |U_{n-1}|$; if $S_{n-2} \in U_{n-1}$ the outer factor is absent and $|U'_{n-1}| = |U_{n-1}| - 1$. In either case $|U'_{n-1}| \le |U_{n-1}|$, so by Corollary 3 the inner space lies in $\bigcap_{j=1}^{\tau(\mu)-|U'_{n-1}|} E^-_j|_{V^\mu} \subseteq \bigcap_{j=1}^{\tau(\mu)-|U_{n-1}|} E^-_j|_{V^\mu}$, i.e. its support has $p \ge \tau(\mu) - |U_{n-1}| + 1 = m - |U_{n-1}|$. Prefix preservation through Φ and through the outer $(T_{n-2}+1)$ when present keeps $p \ge m - |U_{n-1}|$ on V^λ; by the support rule the image lies in $\bigcap_{j=1}^{m-1-|U_{n-1}|} E^-_j = \bigcap_{j=1}^{\tau-|U_{n-1}|} E^-_j$. Finally $|U_{n-1}| \le u$. $\square$

Lemma 4 already establishes (2) for *every* U that keeps T_{n-1}, and shows the deficit is charged *only* to the first–block deletions U_{n-1} — deletions inside the last block below its head are free, because the head S_{n-1} alone forces the image into $\Phi(V^\mu)$. This is the structural reason the empirical kill–point census found interior last–block deletions to be “cost–free”.

Case B1: the leftmost tail generator T_{n-1} is deleted ($g \in U_n$)

Now $\Omega^{(\lambda),[\widehat{U}]}_{\text{tail}}$ contains no T_{n-1}, so it lies in $H_q(S_{n-1})$ and acts block–diagonally on

$$V^\lambda{\downarrow}S_{n-1} \;\cong\; V^{(2,1^{m+1})} \;\oplus\; V^{(2,2,1^{m-1})},$$

the two summands obtained by deleting a corner of λ (the corner $(2,2)$ gives the larger hook $(2,1^{m+1})$; the corner $(m+2,1)$ gives the smaller off–hook $(2,2,1^{m-1})$), each with multiplicity 1. For $j \le n-2$ the generator S_j preserves this decomposition, so $E_j^-|_{V^\lambda} = E_j^-|_{V^{(2,1^{m+1})}} \oplus E_j^-|_{V^{(2,2,1^{m-1})}}$; since the target depth $\tau - u \le \tau = m-1 \le n-2$ involves only such j, it suffices to prove the depth bound on each summand. Write ν for a summand and $C := \Omega_{\text{tail}}^{(\lambda),[\widehat{U}]}|_{V^\nu}$. As $g \in U_n$, the second block has lost its head, so

$$C = \big({R'}_{n-1}\big)^{[\widehat{U_{n-1}}]} \cdot \big({R'}_{n-1}\big)^{[\widehat{W}]}\Big|_{V^\nu}, \qquad W := U_n \setminus \{g\},$$

a product of two gapped copies of ${R'}_{n-1}$, both in $H_q(S_{n-1})$ (generators $\le n-2$), with a combined deletion budget $|U_{n-1}| + |W| = u-1$.

The larger hook summand $\nu = (2,1^{m+1})$. A hook again, so by Lemma 1 *every* generator S_i ($i \le n-2$) is rank one on V^ν, with image supported on arm values $\{i, i+1\}$. Therefore C, a product of rank–one operators, has image pinned by its overall leftmost surviving factor $S_{a^\star}$, where $a^\star = \max\big(\{1,\dots,n-2\} \setminus (\text{letters of } U_{n-1})\big)$ is the head of the first block after its deletions. Exactly as in Proposition 2, the image support has arm value $\ge a^\star$, so $p \ge a^\star - 1$. Since the first block ${R'}_{n-1}$ has $n-2 = m+2$ letters and $|U_{n-1}|$ are deleted, $a^\star \ge (m+2) - |U_{n-1}| \ge (m+2)-(u-1) = m-u+3$. Hence $p \ge m-u+2$ and the image lies in $\bigcap_{j=1}^{m-u+1} E_j^-|_{V^\nu} \supseteq \bigcap_{j=1}^{\tau-u} E_j^-|_{V^\nu}$ (as $\tau - u = m-1-u \le m-u+1$). The bound holds on this summand with room to spare.

The smaller off–hook summand $\nu = (2,2,1^{m-1})$. Here $\tau(\nu) = m-2$. Using ${R'}_{n-1} = S_{n-2}\,{R'}_{n-2}$, rewrite the leftmost block when its head survives: if $S_{n-2} \notin U_{n-1}$,

$$C = S_{n-2} \cdot \Big[\big({R'}_{n-2}\big)^{[\widehat{U'_{n-1}}]}\big({R'}_{n-1}\big)^{[\widehat{W}]}\Big]\Big|_{V^\nu},$$

and the bracket is precisely a two–block tail chain of the shape $(2,2,1^{m-1})$ — whose own Pillar 1 tail is ${R'}_{(n-1)-1}{R'}_{n-1} = {R'}_{n-2}{R'}_{n-1}$ — carrying the deletion set $U'_{n-1} \sqcup W$ of size $\le u-1$. By the induction hypothesis (the deficit law (2) for the smaller off–hook, base $m=2$ where $\tau = 1$ forces $u = 0$ and the statement is plain Pillar 1), the bracket maps V^ν into $\bigcap_{j=1}^{\tau(\nu)-(u-1)} E_j^-|_{V^\nu}$, i.e. its support has $p \ge \tau(\nu) - (u-1) + 1 = (m-2)-(u-1)+1 = m-u$. The outer $S_{n-2} = T_{n-2}+1$, the top generator of the parabolic, preserves the column–1 prefix on V^ν (outer–factor preservation), so the image of C has $p \ge m-u$, landing in $\bigcap_{j=1}^{m-u-1} E_j^- = \bigcap_{j=1}^{\tau-u} E_j^-|_{V^\nu}$. If instead $S_{n-2} \in U_{n-1}$ (the smaller block's head is itself deleted), then $C|_\nu$ contains no S_{n-2} either and branches once more on $V^\nu{\downarrow}S_{n-2}$: the head/branching dichotomy of this section reapplies to the strictly smaller shape ν, a strong induction on m in which the partition shrinks at each descent. This terminates at the base $m=2$.

Theorem 5 (Off–hook deficit Strong Pillar 1)**.** *For $\lambda = (2,2,1^m)$ ($m \ge 2$, $\tau = m-1$) and every $U \subseteq$ tail with $|U| < \tau$,*

$$\Omega_{\text{tail}}^{(\lambda),[\widehat{U}]}\,V^\lambda \;\subseteq\; \bigcap_{j=1}^{\tau-|U|} E_j^-|_{V^\lambda}.$$

Proof. Strong induction on m. The base $m=2$ has $\tau = 1$, so $|U| < \tau$ forces $U = \varnothing$ and the statement is Pillar 1 ($\Omega_{\text{tail}}^{(\lambda)}V^\lambda \subseteq E_1^-$, proved). For $m \ge 3$, split U at the head T_{n-1} of the last block: if T_{n-1} survives, Lemma 4 gives the bound; if T_{n-1} is deleted, the Case B1 analysis reduces the bound, summand by summand, to Lemma 1 on the larger hook $(2,1^{m+1})$ and to the induction hypothesis on the smaller off–hook $(2,2,1^{m-1})$. □

4 Consequences

Corollary 6 (Refined per–subset Pillar 1, uniform in m)**.** *For $\lambda = (2,2,1^m)$ and every tail subset S_T with $|S_T| < \tau$,*

$$M_T(S_T)\,V^\lambda \;\subseteq\; \bigcap_{j=1}^{\tau-|S_T|} E_j^-\big|_{V^\lambda}.$$

Proof. Expand $(q+1)^{|\mathrm{tail}|}M_T(S_T)$ by the identity (1) as a signed sum of $\Omega_{\mathrm{tail}}^{(\lambda),[\widehat{U}]}$ over $U \subseteq S_T$, each with $|U| \le |S_T| < \tau$. By Theorem 5, every term maps V^λ into $\bigcap_{j=1}^{\tau-|U|} E_j^- \supseteq \bigcap_{j=1}^{\tau-|S_T|} E_j^-$. The sum of subspaces of a common $\bigcap_{j=1}^{\tau-|S_T|} E_j^-$ stays inside it. $\square$

This closes the principal gap left open in the May 28 reduction paper, which proved Strong Pillar 1 structurally for only 2 of the 11 single–factor tail positions, and only at $m = 3$. Theorem 5 proves all positions, all multiplicities $|U| < \tau$, and all m at once.

Remark 7 (Toward the full lower bound)**.** Corollary 6 supplies the tail half of the strong vanishing $M(S) \equiv 0$ for $|S| < D$. Writing $M(S) = M_R(S_R)M_T(S_T)$, the depth–$(\tau - |S_T|)$ containment leaves $\tau - |S_T|$ sign–kill layers for the prefix $M_R(S_R)$ to dispatch; combined with a prefix kill (the leftmost $P_q^{(1)}$ on E_1^-, or its deeper analogues) this yields $M(S) \equiv 0$ whenever the prefix can deliver a $P_q^{(j)}$ to one of the surviving layers $j \le \tau - |S_T|$. The remaining content of the full off–hook order law is therefore (i) the prefix counterpart of Theorem 5, controlling $M_R(S_R)$, and (ii) the regime $|S_T| \ge \tau$, both of which lie outside the scope of the present deficit law.

5 Verification

All checks are exact (rational arithmetic at $q = 5/7$, cross–checked at $q = 2/3$), reusing the Hoefsmit seminormal machinery (`compute_p_lambda.py`); scripts in `~/projects/scratch/2026-05-29-strong-pillar1/`.

- *Lemma 1*: for the hook $(2,1^m)$, $m = 3,4,5$, every S_i has rank 1 with image arm–support $\{i, i+1\}$ (and $\{2\}$ for $i = 1$). Exact, both q.
- *Proposition 2/Corollary 3*: for all $A \subseteq \{1,\dots,m+1\}$, $m \le 5$, the nonzero images of Π_A realise depth $\ge \max A - 2$, with the minimum $\tau(\mu) - u$ attained by the leftmost–u deletion.
- *Theorem 5*: the deficit bound $\Omega_{\mathrm{tail}}^{(\lambda),[\widehat{U}]}V^\lambda \subseteq \bigcap_{j=1}^{\tau-|U|} E_j^-$ holds for *every* U with $|U| < \tau$ — exhaustively over all $\binom{2n-3}{u}$ tail subsets for $m = 3$ ($\tau = 2$, 11 tail letters) and $m = 4$ ($\tau = 3$, 13 tail letters); no violation.

λ	m	τ	$\dim V^\lambda$	full–tail depth	min depth over $\lvert U\rvert = 1$
$(2,2,1,1)$	2	1	9	1	— $(\tau{-}1{=}0)$
$(2,2,1,1,1)$	3	2	14	2	$1 = \tau - 1$
$(2,2,1,1,1,1)$	4	3	20	3	$2 = \tau - 1$
$(2,2,1^5)$	5	4	27	4	$3 = \tau - 1$

6 Status

Proved (uniformly in m).

- Lemma 1: every generator S_i is rank one on the hook $(2,1^m)$, image supported on arm values $\{i,i+1\}$ — a self–contained seminormal computation.
- Proposition 2, Corollary 3: the sharp hook deficit bound, with *no* induction (the leftmost factor pins the image).
- Theorem 5: the off–hook deficit law for all U with $|U|<\tau$. Cases A/B2 (the leftmost tail generator T_{n-1} retained) are closed directly by pull–through to the hook; case B1 (T_{n-1} deleted) by S_{n-1}–branching, with the larger hook summand closed by Lemma 1 and the smaller off–hook summand by strong induction on m.
- Corollary 6: refined per–subset Pillar 1 at all sub–threshold levels and all m — the headline application, via the May 28 identity.

Reliances and limits.

- The off–hook lift uses the established R'_n–branching reduction (single hook contributor, no vanishers) and the factor–by–factor pull–through; these are taken from the May 15 meta paper and the May 28 reduction paper, here specialised to $\widehat{\lambda}=(2,2)$.
- In case B1, the smaller off–hook summand is closed by the induction whose descent is the head/branching dichotomy itself; the chain of descents is verified to terminate and to preserve the deletion budget, and the resulting bound is confirmed exhaustively for $m\leq 4$. The bookkeeping of the squaring rewrite $R'_{n-1}=S_{n-2}R'_{n-2}$ across the branching is the same mechanism as the proved May 28 "position 15" lemma, applied recursively.
- The full strong vanishing $M(S)\equiv 0$ for $|S|<D$ is *not* closed here: it additionally needs the prefix counterpart of the deficit law and the regime $|S_T|\geq\tau$ (see the Remark in §4).